



## FALL 2025 - MATH 339: LINEAR ALGEBRA

**Instructor:** Dr. Rachel Barber

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**Office Phone:** 1-301-696-3730

\*\*\*If you email me, please include in the subject of your email "MATH 339."

\*\*\*Available via email 9:00 AM to 5:00 PM M-F. Messages usually returned within 2 business days.

Office Hours: Tues & Thurs: 10:00 AM – 12:00 PM

Fri: 10:00 AM – 11:00 AM

Other office hours may be scheduled by appointment.

### 1. COURSE DESCRIPTION AND OBJECTIVES

A modern introduction to linear algebra and its applications, with an emphasis on geometric interpretation and extensive use of the computer. Topics include: linear systems, matrices, linear transformations, eigenvalues and eigenvectors, diagonalization, and dynamical systems.

**Student Learning Outcomes:** By the end of the course students will be able to:

- (1) Solve systems of linear equations using several different methods, including via a computer.
- (2) Understand the concepts of span and linear independence and dependence, both in  $R^n$  and in more general vector spaces.
- (3) Analyze vector spaces and their subspaces.
- (4) Compute and interpret determinants.
- (5) Understand the difference between consistent and inconsistent linear systems and be able to describe them geometrically.
- (6) Find the eigenvalues and eigenvectors of a square matrix and explain their significance.
- (7) Diagonalize matrices.
- (8) Write and speak effectively about mathematics.

**Textbook:** *Linear Algebra and Its Applications, 6th Edition* by David C. Lay. Pearson, 2016.

**Calculating Tools:** Learning calculus today includes using computers as calculation aids, similar to how calculators were used earlier. While no special calculator is required, a scientific calculator is recommended for this course.

These are some of the tools we may use during the course:

- **Excel:** Available in all campus labs and through Hood's Office 365.
- **Desmos** (<https://desmos.com>): A free graphing calculator with some calculus features. Apps available for iOS and Android.
- **Maple:** A powerful computer algebra system installed on Hood lab computers. Offers advanced features and helps keep a record of your work. A student license can be purchased but is not required.

**Class Meeting Information:** Mon & Wed • 11:30 AM – 12:55 PM in Hodson 308G.

A calculator is recommended for this course.

## 2. GRADING AND COURSE EVALUATION

| Grading Distribution:       |      | Grading Scale:          |                         |
|-----------------------------|------|-------------------------|-------------------------|
| Homework                    | 20%  | <b>A:...</b> 100% – 93% | <b>C:...</b> 76% – 74%  |
| Quizzes/In-Class Activities | 15%  | <b>A-:...</b> 92% – 90% | <b>C-:...</b> 73% – 70% |
| Project                     | 10%  | <b>B+:...</b> 89% – 87% | <b>D+:...</b> 69% – 67% |
| Midterm Exams               | 30%  | <b>B:...</b> 86% – 84%  | <b>D:...</b> 66% – 64%  |
| Final Exam                  | 25%  | <b>B-:...</b> 83% – 80% | <b>D-:...</b> 63% – 60% |
| Total                       | 100% | <b>C+:...</b> 79% – 77% | <b>F:...</b> 59% – 00%  |

*Grades are rounded to the nearest whole number at the end of the semester according to mathematical rules: if the decimal value to the right is 5 or greater, then it will be rounded up, if it is less than 5, then it is rounded down.*

**Attendance.** Attendance is not graded, but regular participation is essential for success. Frequent absences may affect your ability to pass. **If you must miss a class, please notify me promptly.** Be sure to get notes from classmates and check Blackboard for updates.

**Homework.** Homework reinforces what we learn in class. Assignments are given when a topic/section is finished and due a few days later. There are two types: computational problems on WeBWorK and more challenging problems to submit on paper or Blackboard. Collaboration is encouraged, but submit your own work and credit any collaborators. **Homework must be submitted by the assigned due dates.**

**Math Lectures and Seminars.** You are required to attend one math-related event outside class for each upper-level math course you take. This will be recorded as a homework grade. I will announce upcoming activities throughout the semester.

**Quizzes and In-Class Activities.** Quizzes and group activities happen throughout the semester. Use your textbook and notes during quizzes. In-class activities support learning but are usually ungraded. Quizzes count for 15% of your final grade.

**Project.** Mid-semester, you will begin working in teams on a project chosen from a provided list or approved by me. The final presentations will take place during the last one or two class sessions. More details will be available on Blackboard. The project counts for 10% of your final grade.

### Exams

There will be two in-class exams, each worth 15%, plus a comprehensive final exam worth 25%. Missing an exam results in a zero for that exam.

**Tentative exam dates:**

**Exam 1:** Week of September 15, 2025

**Exam 2:** Week of October 20, 2025

**Final Exam:** See Hood College final exam schedule.

### Late Work/Make-Up Policy

Late homework and make-up work may be accepted case-by-case, but no late work will be accepted once graded assignments are returned. Extensions require mutual agreement on a new deadline. **No late or make-up work will be accepted after the official last day of classes as listed in the academic calendar.**

**Extra Credit:** Extra credit may be given out solely at the instructor's discretion.

## 3. CLASS POLICIES

**Electronic Devices Policy:** You're responsible for your own learning. Misuse of devices won't be policed but may hurt your understanding and grades. If you need to use a device for an emergency or non-class reason, please step outside to avoid disrupting others.

**AI and Unauthorized Aid Policy:** You may use AI and other resources to help with homework, but don't rely on them to do the work for you. Understanding the material yourself is essential for success on exams. Collaboration on homework, projects, and quizzes is encouraged, but submitted work must be your own.

- **Homework:** Discuss and work with classmates, but ensure your submitted work reflects your own understanding.
- **Quizzes and Projects:** Discuss and work with classmates, use your textbook and notes. AI and internet/technology resources allowed will be directly specified on the assignment or in-class.
- **Tests/Exams:** No outside help allowed, including AI, internet, textbooks, notes, or peers. Exams must reflect your individual knowledge.

Violations risk failing the assignment or course and academic discipline. If unsure about allowed resources, ask me.

#### 4. COLLEGE INFORMATION

**Hood Academic Honor Code:** As a place of honor and respect, all members of Hood College assume the obligation to maintain the principles of honesty, responsibility, and intellectual integrity in all activities related to their Hood College experience. Students are expected to adhere to the highest standards of academic honesty and integrity in all coursework and related matters. It is the responsibility of each student to support these values through maturity of thought, expression, and action. Members of the faculty and staff are available to assist students in this process. The Academic Judicial Council, which is comprised of undergraduate students and two elected faculty members, is chaired by Dean Paige Eager. If you have questions about the Academic Judicial Council or the Hood College Honor Code, please contact Dean Eager at [eager@hood.edu](mailto:eager@hood.edu). All Honor Code matters related to graduate courses and graduate students should be referred to Dean April Boulton at [boulton@hood.edu](mailto:boulton@hood.edu).

**Support Services:** If you're feeling stressed or notice someone struggling, please consider reaching out for help. Amanda Dymek's office and Hood Counseling Services offer support. Counseling is available M–F, 8:30 AM–5:00 PM, Apple Resource Building, 1st floor, Room A. You can make appointments online at <https://hood.edu/counseling> or drop in for walk-ins on Mon, Wed, and Fri from 11:00 AM–12:00 PM. Questions? Email [counselingservices@hood.edu](mailto:counselingservices@hood.edu).

Other mental health resources include the NeighborHood Counseling Training Center and Thriving Campus.

For immediate support anytime, text or call **9-8-8** for confidential counseling 24/7.

**Accessibility Services:** This course is designed to be accessible to all students, including those with disabilities or conditions affecting access. If you experience any barriers, please contact me or the Office of Accessibility Services at 301-696-3569 or [accessibilityservices@hood.edu](mailto:accessibilityservices@hood.edu).

If you have prior accommodations, Accessibility Services can help you create a plan to ensure support without requiring disclosure to instructors.

**Title IX Information:** Hood College faculty are committed to helping create a safe and open learning environment for all students. If you (or someone you know) have experienced any form of sexual misconduct, including sexual assault, dating or domestic violence, or stalking, know that help and support are available. The College strongly encourages all members of the community to take action, seek support, and report incidents of sexual misconduct to the Title IX Coordinator. Please be aware that under Title IX of the Education Amendments of 1972, I am required to disclose information about such misconduct to the Title IX office. If you wish to speak to a confidential employee who does not have this reporting responsibility, you can contact the Dean of the Chapel ([chapel@hood.edu](mailto:chapel@hood.edu)), counseling services ([counselingservices@hood.edu](mailto:counselingservices@hood.edu)), or health services ([healthservices@hood.edu](mailto:healthservices@hood.edu)). For more information about reporting options and resources at Hood College and the community, please contact the Title IX Coordinator.

**Hood's commitment to inclusivity and campus values:** Hood College is proud of its diverse community, and we are committed to cultivating and strengthening an inclusive, tolerant, multi-cultural, and intellectually open community with equal opportunity for all. By encouraging and celebrating our differences, we create an environment that promotes freedom of thought and academic excellence. It is our goal to have a respectful and nurturing academic community that

affirms the inherent worth and dignity of every individual, and celebrates the diverse backgrounds of all students, faculty, and staff. We will strive to value each person for their uniqueness and difference and to encourage all community members to reach their fullest potential.

**Land Acknowledgement.** Hood College acknowledges that it was established on lands in harmony with tribes of the Algonquian and Iroquois language families. Many tribes, including the Piscataway, Susquehannock, and Tuscarora tribes, passed through and resided in this region. Hood College recognizes this past and offers humble respect to the indigenous peoples of the past, present, and future connected to this place.

**Student Success Services:** Your success is important to me. I use Navigate 360 to identify and support students who may be struggling with attendance, participation, or performance. If needed, I may send an Academic Alert to connect you with campus resources. These alerts are meant to help, not punish.

Please respond promptly to any communications from me, your advisor, or campus offices regarding your progress.

**Rachel Barber reserves the right to revise this syllabus at any time as necessary during the semester. Any changes will be communicated promptly to all students, and updated information will be made available through the appropriate channels.**